Sleep Health & Lifestyle Analysis

Uncovering insights into sleeping patterns, physical activity, and the relationships between stress, BMI, and sleep quality — using Python, SQL, and Power BI.

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Project Overview

Dataset

374 individuals across various occupations

Tools

Python, SQL, and Power BI

Goal

Guide people toward healthier lifestyles through data-driven insights

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Dataset Summary

Structure

374 rows × 13 columns — no missing or duplicate values

Person Details

Age, Gender, Occupation, BMI Category

Lifestyle Behavior

Sleep Duration, Quality, Physical Activity, Stress Level, Daily Steps, Sleep Disorder

Medical Parameters

Blood Pressure, Heart Rate

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Exploratory Data Analysis — Python

A structured Python workflow ensured data quality before analysis — from loading and exploration to standardization and database integration.

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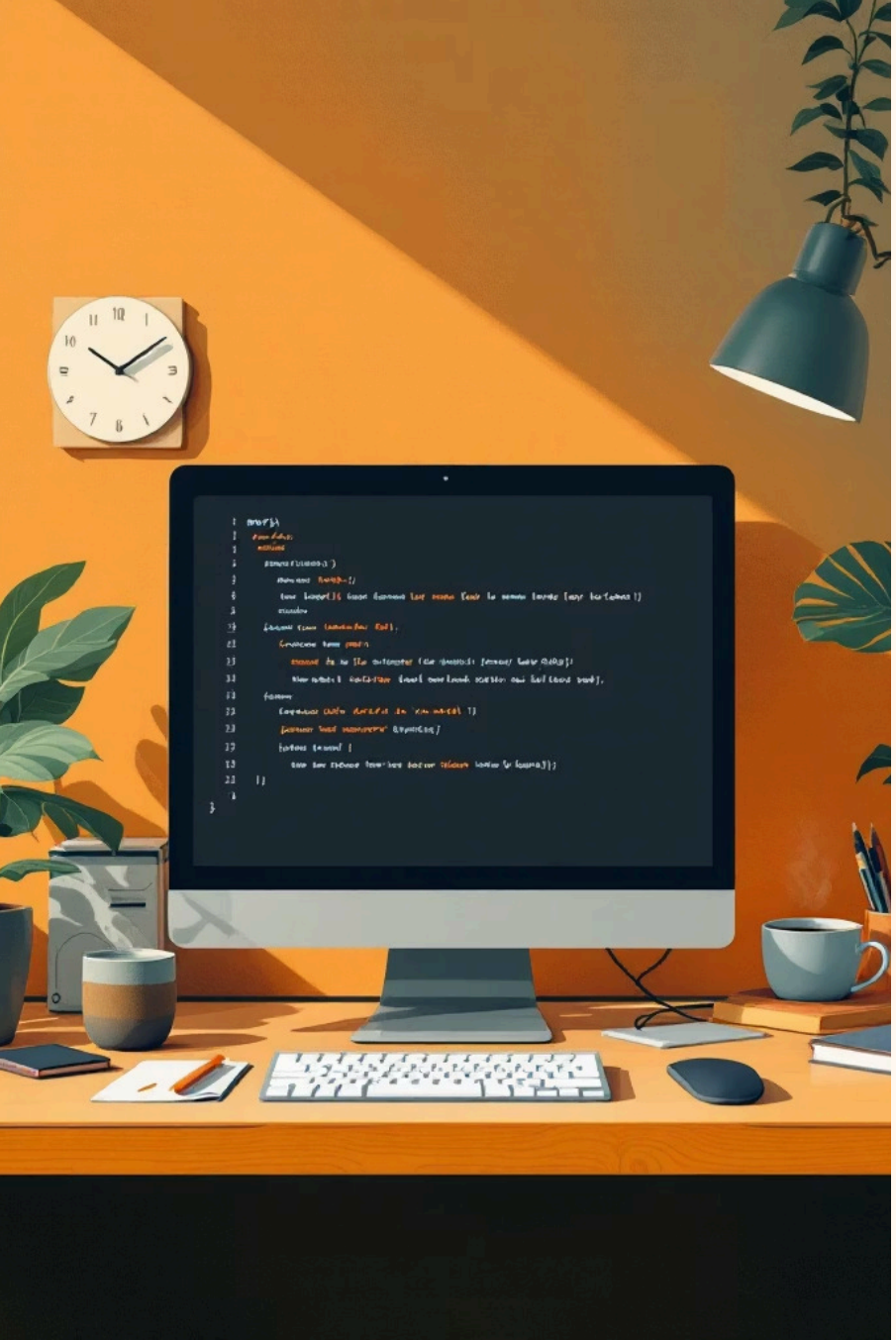

Data Analysis Using SQL

Structured queries in MySQL answered key questions about stress, activity, and BMI across occupations.

1

Top 3 Occupations

Highest average stress levels

2

Top 5 Most Active

Ranked by daily steps

3

BMI vs. Stress

Stress levels across BMI categories

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SQL Findings — Stress & Activity

Stress by Occupation

Top 3 occupations with the highest average stress levels were identified through SQL analysis.

Daily Steps Ranking

Top 5 most active individuals ranked by daily steps.

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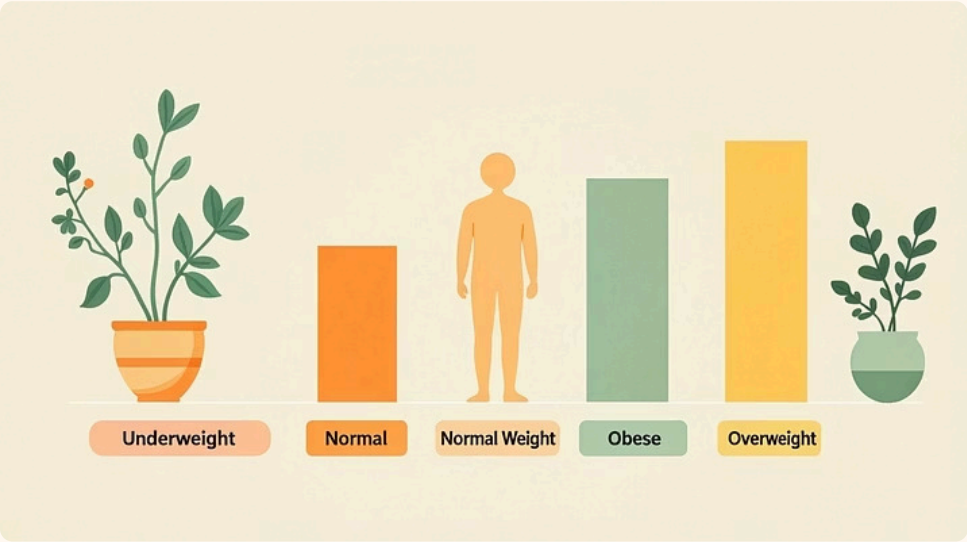
SQL Findings — BMI & Stress



BMI Category vs. Stress Level

Analysis revealed that **overweight individuals** experience higher stress levels, while those in the **normal BMI** category report lower stress.

Stress and BMI are closely linked — a key insight for lifestyle intervention strategies.

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Power BI Dashboard

An interactive dashboard with KPIs and visual insights for quick, actionable overviews.

6.82K

Avg Steps

7.13

Sleep Duration (hrs)

5.39

Avg Stress Level

7.31

Sleep Quality Score

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Key Insights — Sleep & Demographics

Stress & Sleep Duration

People with lower stress levels sleep **more than 7 hours** on average.

Gender Differences

Females have higher average sleep duration; males report higher average stress levels.

Sleep Disorders

Those with no sleep disorder sleep **7+ hours**; people with insomnia sleep the least.

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Conclusion

Lifestyle factors — physical activity, stress, and BMI — **significantly impact sleep health.**

🏃 Activity

Nurses take the most daily steps; Sales Representatives the fewest.

⚖️ BMI & Stress

Overweight individuals report higher stress; normal BMI individuals report less.

📊 Actionable Insights

The dashboard provides clear guidance for improving overall well-being.

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